

MASHROOR HASAN BHUIYAN

PhD Student, Department of Computer Science, The University of Texas at Dallas

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EDUCATION

PhD in Computer Science

August 2025 – July 2030

The University of Texas at Dallas, Richardson, TX, USA

CGPA: 3.75/4.00

BSc in Computer Science and Engineering

February 2020 – March 2025

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

CGPA: 3.94/4.00

RESEARCH INTERESTS

Security and Privacy of Cellular and NextG Systems (LTE/5G, O-RAN, Voice over Wi-Fi) — Protocol Security Testing and Specification Analysis — ML-Based Network Intrusion Detection and Explainable Mitigation — LLM-Powered Cloud Security Analytics

RESEARCH POSITIONS

The University of Texas at Dallas

Richardson, TX, USA

Graduate Research Assistant

August 2025 – August 2026

Department of Computer Science, Erik Jonsson School of Engineering

Advisor: Dr. Imtiaz Karim

Inverse AI

Dhaka, Bangladesh

Machine Learning Engineer

April 2025 – July 2025

Research & Development

RESEARCH PUBLICATIONS

Imtiaz Karim, Hyunwoo Lee, Hassan Asghar, Kazi Samin Mubasshir, Seulgi Han, Mashroor Hasan Bhuiyan, and Elisa Bertino, “*VWAttacker: A Systematic Security Testing Framework for Voice over WiFi User Equipments*” The 45th IEEE International Conference on Computer Communications (**INFOCOM**), 2026

MANUSCRIPTS UNDER REVIEW

Mashroor Hasan Bhuiyan, Indranil Biswas, Fatema Tuj Johora, and Imtiaz Karim, “*5GAgent: Agentic Security Testing of 5G Baseband Control-Plane Implementations*”

The 34th Network and Distributed System Security Symposium (**NDSS**), 2027

PROJECTS

GSMA Open Telco AI, Open Data Contributor

September 2025 – March 2026

Training, benchmarking and fine-tuning of telecom-domain language models. Launched at MWC Barcelona 2026.

huggingface.co/GSMA

Network-Side Fake Base Station Detection

October 2025 – July 2026

Design, implementation and evaluation of FBS detection in real-world vendor networks. In collaboration with Purdue University and Polaris Inc.

FELLOWSHIPS AND AWARDS

CyberPowder Fellow

January 2026 – May 2026

NSF CyberTraining Program (Award #2417934), University of Utah POWDER Platform. Project: *Deploying Autonomous Security Testing of 5G UE Implementations on POWDER*

Dean's List*November 2021 – November 2023*

Bangladesh University of Engineering and Technology (BUET). Awarded in all four years of undergraduate study

University Technical Scholarship*February 2020 – March 2025*

Bangladesh University of Engineering and Technology (BUET). Merit-based, awarded for academic performance

ICPC Asia West Continent Finals*March 2025*

Contestant representing Bangladesh University of Engineering and Technology (BUET). Qualified via ICPC Dhaka Regional

Bangladesh National Team*July 2019*

60th International Mathematical Olympiad, Bath, United Kingdom

WORKSHOPS AND TRAINING

Open6G+AI School*April 2026*

Institute for Intelligent Networked Systems, Northeastern University

First OAI Foundation U.S. Hands-on Workshop*November 2025*

OpenAirInterface Foundation, The University of Texas at Austin

ACADEMIC SERVICES

- Reviewer: IEEE Transactions on Networking, 2025 [External]
- Reviewer: IEEE Transactions on Dependable and Secure Computing (TDSC), 2026 [External]

SKILLS

- Programming: Python, C/C++, Java and SQL
- Machine Learning & Explainability: Deep Learning, Unsupervised Classification, Anomaly Detection, Feature Engineering, Model Quantization, Feature Attribution and Explanation-Guided Decision Support
- Large Language Model Systems: Retrieval-Augmented Generation, Instruction Tuning, Parameter-Efficient Fine-Tuning (LoRA), LLM Agents, Tool Use, Multimodal LLMs and LLM Evaluation
- Cellular & Network Security: LTE/5G Protocol Analysis, Voice over Wi-Fi Security, O-RAN Security, 3GPP Specification Analysis, Stateful Network-Trace Analysis, Threat Modeling and Security Testing